

High Resolution DPWM in a DC-DC Converter Application Using Digital Sigma-Delta Techniques

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Abstract— This paper demonstrates a dc-dc converter application in which a multi-bit digital sigma-delta modulator pre-processes the duty-cycle value before application to the DPWM, increasing the effective resolution of the DPWM dramatically. In this method, the DPWM duty-cycle command is pre-processed by a multi-bit digital sigma-delta modulator, so that the DPWM quantization noise will be shaped in frequency. As a result, the total quantization noise at the output of the dc-dc converter is reduced, and the effective resolution of the DPWM in the control loop is increased dramatically. A prototype converter was seen to perform with an effective DPWM resolution of at least 11 bits, with an actual DPWM resolution of less than 6 bits.

I. INTRODUCTION

The design of the Digital Pulse Width Modulator (DPWM), has been the subject of intensive investigation in the design of digitally controlled dc-dc converters [2], [4-7], [11]. It has been shown, that in order to eliminate limit-cycles at the output of a dc-dc converter, the resolution of the DPWM is critical [1], [3]. Limit cycles have been problematic in published controllers [1] and papers demonstrate the appearance of unwanted limit cycles when the DPWM resolution is too low [14]. All of these DPWM designs are fundamentally similar designs, in that the input word to the DPWM is resolved directly to a pulse width. One author has

taken a different approach, using a dithering technique to increase the effective resolution of the DPWM [3].

This paper will demonstrate a pre-processing approach combined with a standard, counter based DPWM, to achieve very high effective resolution. In this method, the DPWM command is pre-processed by a multi-bit digital sigma-delta modulator, so that the DPWM quantization noise will be shaped in frequency, reducing the noise at lower frequencies in exchange for more noise at higher frequencies, which will then be attenuated by the dc-dc converter's LC filter. As a result, the total quantization noise at the output of the dc-dc converter is reduced, and the effective resolution of the DPWM in the control loop is increased dramatically. This is a low complexity solution when compared to previous DPWM solutions, and the sigma-delta does not affect the stability of the closed loop, because its signal transfer function is unity magnitude, and delay free, leaving the closed-loop pole placements undisturbed compared to a standard DPWM solution.

Analog sigma-delta modulators have been considered as a means of reducing the wide-band electro-magnetic interference (EMI) which is inherent in PWM [8,9]. The approach has been to replace an analog PWM with a single-

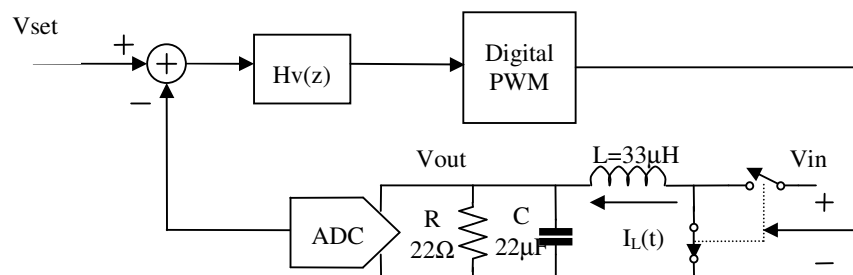


Figure 1. Prototype buck converter with sigma-delta noise shaped DPWM

bit sigma-delta modulator in a traditional analog dc-dc converter.

This authors' design involves a multi-bit digital sigma-delta modulator for noise shaping. Multi-bit sigma-delta modulators have been used with success in audio signal processing applications [10,11].

The contribution of this paper is to demonstrate that high performance dc-dc conversion, which depends on a high-resolution DPWM, can be achieved by a very simple, low resolution DPWM with multi-bit digital sigma-delta pre-processing. In this example, greater than 13 bit performance is achieved from a 6 bit DPWM.

II. DC-DC CONVERTER

A typical digitally controlled dc-dc converter is shown in Figure 1, as a buck configuration. It is a voltage-mode controller, comprising of a digital controller H_v , a digital PWM, power-train and ADC.

In order to eliminate limit-cycles at the output of the converter, the voltage resolution of the DPWM is critical [1], [3]; it must be better than the voltage resolution of the ADC in order for the DPWM to resolve to a voltage corresponding to zero steady-state control error. Typical ADC resolution is 8 to 10 bits of the input voltage range, which corresponds to approximately 10 – 100mV.

III. DIGITAL PWM

The resolution of a DPWM is limited by the discrete values which it may resolve. Generally, the DPWM duty cycle is quantized to multiples of smallest clock period. An effective, but high-complexity, technique is to resolve to divisions of the smallest clock period using a delay-locked-loop (DLL) [6,7].

The quantization inherent in the limited resolution of the DPWM can be represented, in a linear model, as noise ϵ ,

being added to the non-quantized value to produce the quantized result, which is applied to a linear PWM, Figure 2.

Looking at the spectrum of the quantization noise, Figure 4 (b), we can see that it tends to be white; and therefore, the quantization, or (equivalently) resolution, of the DPWM applies equally to all frequencies, up to the Nyquist rate of the sampled data system.

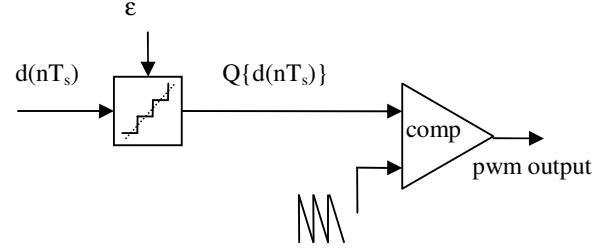


Figure 2. Linear model of a DPWM as a PWM preceded by the addition of quantization noise

Examining how the DPWM quantization noise contributes to the dc-dc converter output noise, Figure 3, we can express the output-noise contribution as:

$$S_o(\omega) = S_{dpwm}(\omega) |H_p(e^{j\omega})|^2 |H_{rej}(e^{j\omega})|^2 \quad (1)$$

, where $S_o(\omega)$ is the output noise spectral density, $S_{dpwm}(\omega)$ is the quantization noise spectral density of the DPWM, H_p is the transfer function of the power-train and H_{rej} is the transfer function for closed-loop rejection of disturbances at the output.

Figure 4 (a), shows the gain magnitude of the power stage of Figure 1, and the power spectral density of the DPWM quantization noise.

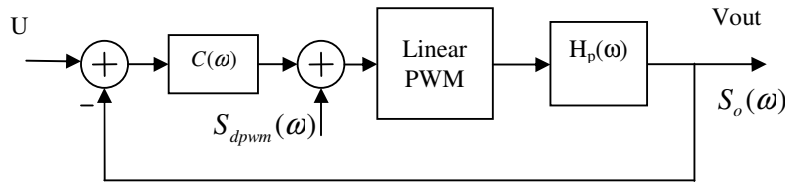
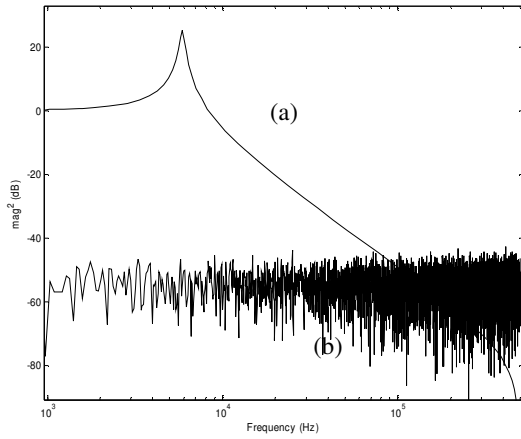
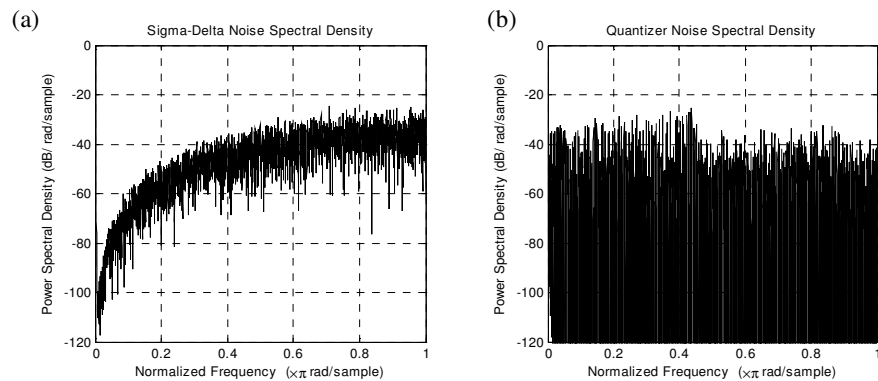


Figure 3. Model of the dc-dc converter with quantization noise



**Figure 4. (a) Plant Magnitude;
(b) Quantization Noise Power Spectral Density;
sample-rate = 1MHz**

With its logarithmic magnitude axis, it's plain from Figure 4, that the quantization noise is attenuated significantly at high frequencies by the power train's LC-filter, and from (1) this reduces the output noise spectral density at high frequencies when compared to rejection at the loop's output without filtering. Thus, increasing the noise at higher frequencies will have little impact on the overall noise power, whilst reduced noise at low frequencies is clearly advantageous. A comparison of the power spectral density of the sigma-delta noise versus the noise from a quantizer of the same resolution, Figure 5, further illustrates the advantages of sigma-delta.

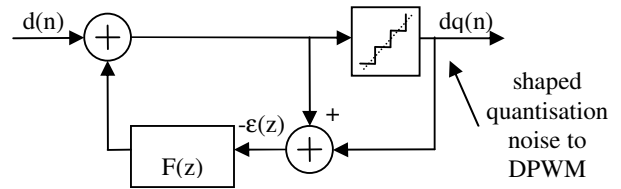


**Figure 5. (a) Sigma-Delta Noise Power Spectral Density, 5-bit quantization (32-levels);
(b) Quantization Noise Power Spectral Density, 5-bit quantization (32-levels)**

IV. MULTI-BIT DIGITAL SIGMA-DELTA MODULATOR

The strategy in this method is to quantize the duty-cycle value $d(n)$, to the resolution of the DPWM, and to shape the resulting quantization noise so that there is less noise at low frequencies, passed by the LC filter; and more noise at high frequencies attenuated by the LC filter. This pre-quantized duty cycle value $dq(n)$, is then applied to the DPWM, and because the resolution of the quantizer in the sigma-delta is matched to that of the DPWM, all values are resolvable to a DPWM level without additional quantization.

Noise shaping may be achieved using a multi-bit digital sigma-delta modulator, as shown in Figure 6, where $F(z)$ is the loop filter. In a first-order error-feedback sigma-delta, $F(z)$ is just a delay of one sample. A second-order error-feedback sigma-delta comprises two taps. The sigma-delta shapes the quantization noise introduced by its quantizer, as shown in Figure 5(a).



**Figure 6. Error feedback sigma-delta modulator,
shaping the noise applied to the DPWM**

Figure 7 shows the quantized duty-cycle value $dq(n)$ in the time domain. Note the high-frequency dithering between quantization values.

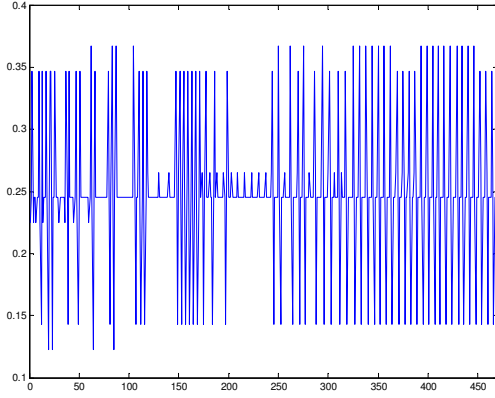


Figure 7. $dq(n)$, showing the dithering to discrete DPWM levels after quantization by the sigma-delta

V. EQUIVALENT RESOLUTION OF THE MULTI-BIT SIGMA-DELTA MODULATOR

Assuming that the LC filter can be approximated by a brick-wall filter with bandwidth f_b , eliminating all noise outside its bandwidth, we can calculate the equivalent signal to noise ratio in dB, for a first-order multi-bit sigma-delta DPWM (2), where f_s is the sample rate, and N is the number of quantization levels [13]:

$$SNR = 5.62 + 20\log_{10}N + 30\log_{10}\left(\frac{f_s}{2f_b}\right) \quad (2)$$

We consider the brick-wall filter assumption to be reasonable because of the very low cutoff frequency of the LC filter compared to the system sample rate, and its 2nd order cutoff characteristics, (refer to the example of Figure 4, and note the Log frequency axis).

The equivalent number of bits (ENOB) can be calculated (3), where SNR is in dB:

$$ENOB = \frac{SNR - 1.76}{6.02} \quad (3)$$

The applicability of equation (2) to the steady state situation deserves some attention. The third term of (2) predicts 9dB of SNR improvement for each doubling of the oversampling ratio. One part of that improvement (6dB) is due to the noise shaping of the sigma-delta loop, and the remaining 3dB is due to the assumption that the noise power is halved if the oversampling ratio is doubled. That assumption however, is predicated on the understanding that the quantization noise power is white, and is therefore evenly distributed across all frequencies. Under steady-state conditions this assumption cannot be justified, and we should therefore view equation (2) as a best case of what can be achieved. It is possible to set a lower bound on the SNR and ENOB by disregarding the improvement due to oversampling the noise, as follows:

$$SNR = 5.62 + 20\log_{10}N + 20\log_{10}\left(\frac{f_s}{2f_b}\right) \quad (4)$$

VI. APPLICATION

The sigma-delta noise shaped DPWM has been applied in a prototype buck converter, Figure 8. The transfer function for the controller is:

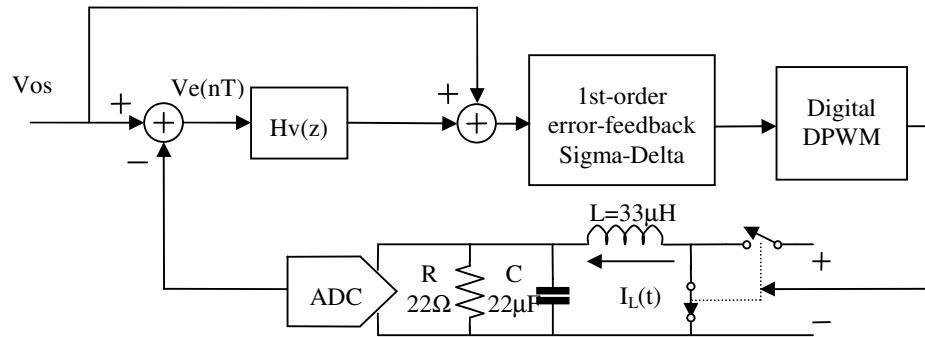


Figure 8. Prototype buck converter with sigma-delta noise shaped DPWM

$$Hv(z) = 55(1 - 0.9z^{-1}) \quad (5)$$

The controller was prototyped on an ADSP21992 DSP prototyping system, which integrates DPWMs and ADCs. The power components were prototyped on an AD1991 power switching IC, developed for class-D applications. The sample rate was $f_s=1\text{MHz}$; as was the switching frequency of the DPWM. The particulars of clock generation in the prototype, gave a DPWM resolution of $1/50$, which is equivalent to 5.6 bits. The ADC resolution was 11 bits.

Taking f_b as the loop bandwidth; 10kHz in this case, then from (4) and (3) the lower bound of the SNR and effective number of bits is:

$$SNR = 73.6\text{dB} \quad (6)$$

$$ENOB = 11.9 \text{ bits} \quad (7)$$

The upper bound on the ENOB is 14bits.

Therefore, in this application the DPWM operates at 12bit resolution, at least. This is equivalent to a master clock rate of 4096MHz for a counter based DPWM, with a 1MHz pulse rate. The closed loop pole placement of the loop is undisturbed by the inclusion of the sigma-delta because its signal transfer function is unity gain and delay-free. Therefore the dynamics of the loop are unchanged by the sigma-delta modulator.

VII. RESULTS

The performance of the prototype for a positive load-step is shown Figure 9. It is evident from the structure of the steady state response, that there is no periodic limit cycling, indicating that the effective resolution of the DPWM was higher than that of the ADC (11 bits). There is no evidence that the DPWM resolution was only 200mV ($10\text{V}/50$); any noise in the steady state appears random.

VIII. CONCLUSION

It has been shown, in a dc-dc converter application that pre-processing the duty-cycle value using a multi-bit digital sigma-delta modulator, before the DPWM increases the effective resolution of the DPWM dramatically. A prototype converter was seen to perform with an effective DPWM resolution of at least 11 bits, with an actual DPWM resolution of less than 6 bits.

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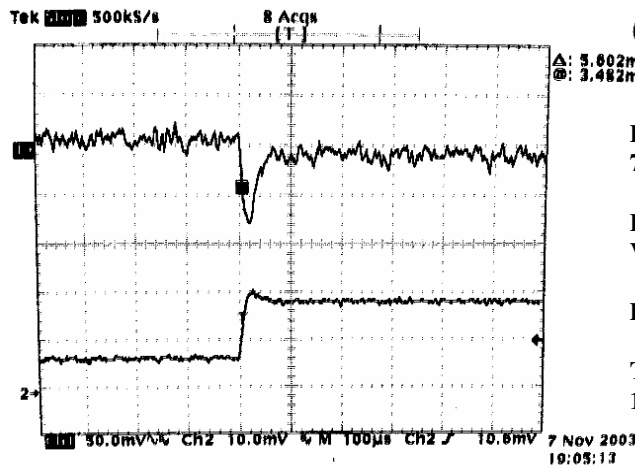


Figure 9 . Experimental 25% - 75% positive load step;

$L=33\mu\text{H}$, $C=22\mu\text{F}$, $F_s=1\text{MHz}$,
 $V_{in}=10\text{V}$, $V_o=2.5\text{V}$,

PWM resolution = $1/50$ (200mV).

Trace1: $50\text{mV}/\text{div}$ vertical scale,
 $100\mu\text{s}/\text{div}$ horizontal scale.

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